

INSTRUCTIONS: You may use any computer language or mathematical software for this project. Submit well-commented source code or computer algebra system code along with your output. You may submit the project in multiple files. DO NOT submit the zipped files. Project should be typed. **Any handwritten part may not be graded.** If you are submitting your code and the output in other format, please also include a copy in pdf format. **Include the names of all students participated in the project at the top of the files.**

Suppose that you would like to send the following message to your instructor in a secure way using AES cipher:

I need an A+ plz

which in hexadecimal notation:

49 20 6E 65 65 64 20 61 6E 20 41 2B 20 70 6C 7A

1. You selected the key:

45 6E 63 72 79 70 74 20 6D 79 20 6D 61 72 6B 73

Show the round key K_0 as a 4×4 matrix with entries in hexadecimal notation.

2. By using the key schedule given in the class notes slides, find the round key K_1 as a 4×4 matrix with entries in hexadecimal notation.
3. Encrypt your message in the first round of the AES cipher and explicitly show the matrix (with entries in hexadecimal notation) after each step as indicated below:
 - (a) Show the 4×4 matrix after adding the key K_0 .
 - (b) Show the 4×4 matrix after SubBytes step.
 - (c) Show the 4×4 matrix after ShiftRow step.
 - (d) Show the 4×4 matrix after MixColumns step.
 - (e) Show the 4×4 matrix after adding the key K_1 .